

Kren-M: Meghalaya's First Foundational AI Model for the Khasi Language

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Abstract

We present Kren-M, Meghalaya's first foundational AI model and the first generative language model built in Northeast India for an indigenous language. This 2.6-billion-parameter bilingual (Khasi-English) LLM, based on Gemma-2-2B, incorporates a custom tokenizer with 2,135 added tokens, yielding 30–36 % tokenization efficiency gains for Khasi and Garo. Continued pre-training on 5.43 million rigorously cleaned Khasi sentences (proprietary) is followed by instruction tuning on 33,034 high-quality examples. Key technical fixes eliminate auto-translation, instruction echoing, and infinite generation. Kren-M delivers fluent, task-aware bilingual chat and translation. The model and all checkpoints are publicly released on Hugging Face as MWirelabs/Kren-M. To support Northeast Indian language research, we additionally released one of the largest public Assamese and Mizo corpora, along with the first public Garo corpus.

Keywords: Khasi LLM, Meghalaya AI, Northeast India LLM, low-resource language model, Khasi language AI, Indic LLM, foundational model, Low-resource languages, Northeast India AI Model, Multilingual NLP

1. Introduction

Low-resource languages face significant barriers in the modern digital landscape, with limited access to natural language processing technologies that are increasingly essential for digital inclusion (Joshi et al., 2020). Khasi, a Austroasiatic language spoken by approximately 1.4 million people primarily in Meghalaya, India, exemplifies this challenge despite having official recognition and written standardization. To the best of our knowledge, Kren-M is the first generative language model trained and publicly released from Northeast India for an indigenous language.

Recent advances in large language models have demonstrated remarkable capabilities across diverse tasks (Brown et al., 2020; OpenAI, 2023), yet these models predominantly serve high-resource languages. While multilingual models like mBERT and XLM-RoBERTa provide some coverage, their performance on truly low-resource languages remains limited due to vocabulary inefficiencies and insufficient training data (Conneau et al., 2020).

This work introduces Kren-M, a bilingual language model specifically designed for Khasi-English language pairs. We address key technical challenges through: (1) custom vocabulary extension via SentencePiece tokenization, (2) systematic continued pre-training on cleaned monolingual corpora, and (3) instruction tuning with careful attention to common failure modes. Our contributions include publicly releasing the first production-ready Khasi language model and documenting reproducible methods for adapting foundation models to low-resource languages.

2. Related Work

Adapting language models to low-resource languages has been explored through various approaches. Continued pre-training has proven effective for domain and language adaptation (Gururangan et al., 2020), while vocabulary extension methods improve tokenization efficiency (Chung et al., 2020). For Indian languages specifically, efforts like IndicBERT (Kakwani et al., 2020)

and AI4Bharat's models have made significant progress, though Northeast Indian languages remain underserved.

Parameter-efficient fine-tuning methods, particularly LoRA (Hu et al., 2022), have enabled efficient adaptation of large models. Recent work on instruction tuning (Ouyang et al., 2022; Chung et al., 2024) has demonstrated the importance of careful data curation and training techniques for achieving production-ready conversational capabilities.

3. Methodology

3.1. Tokenizer extension

We address vocabulary inefficiency by training a custom SentencePiece tokenizer on 66,188 sentences (60,000 Khasi + 6,188 Garo) with a 5,000 token vocabulary. From this model, we extract 2,135 tokens that are poorly represented in the original Gemma-2-2B tokenizer (requiring ≥ 2 subwords). These tokens are added to the base vocabulary, expanding it from 256,000 to 258,135 tokens. This yields 30-36% tokenization efficiency improvements for Khasi and Garo text while preserving English performance.

Table 1. Tokenization efficiency on held-out 100-sentence test set (average tokens per sentence).

Language	Gemma-2-2B (base)	Kren-M (extended vocab)	Reduction
Khasi	128	82	36%
Garo	109	76	30%
English	89	89	0%

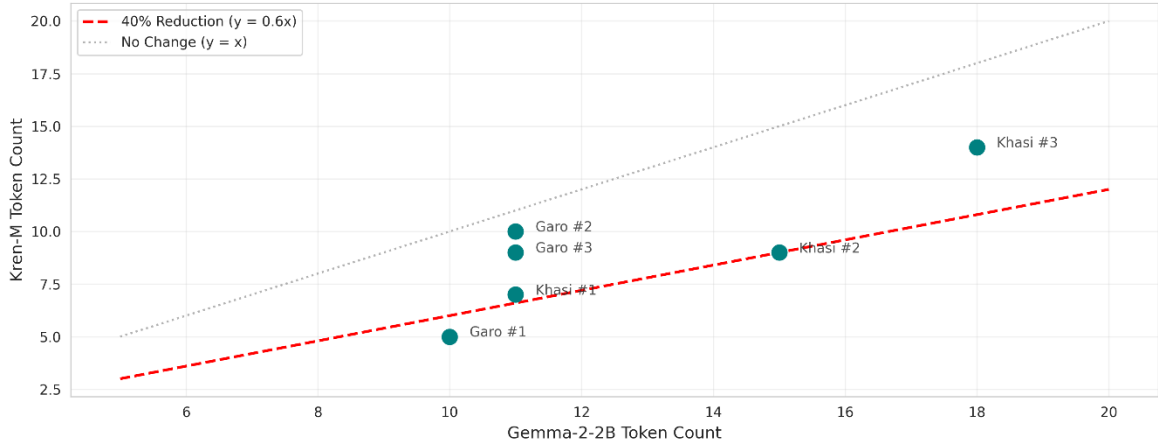


Figure 1. Token efficiency comparison between base Gemma-2-2B and Kren-M across increasing prompt lengths on Khasi text. Kren-M maintains near-linear scaling (slope ≈ 0.6) while the base model degrades steeply.

3.2. Continued Pre-training

Continued pre-training was performed on a proprietary corpus of 5.43 million high-quality Khasi sentences after extensive cleaning that removed 279,247 contaminated sentences (4.9 %). To support future research on related Northeast Indian languages, we publicly released one of the largest Assamese and Mizo corpus and the first publicly available Garo corpus on Hugging Face under MWirelabs/assamese-monolingual-corpus, MWirelabs/mizo-language-corpus-4M, and MWirelabs/garo-english-parallel-corpus respectively.

Training proceeds in two stages: Stage 1 trains for one epoch with learning rate $2e-4$, achieving validation loss of 2.974 (56% reduction from baseline 6.77). Stage 2 applies a refinement pass at $1e-4$ learning rate, reaching 2.941. We use LoRA with rank 32, targeting all projection layers while fully training embedding and language model head layers to accommodate the extended vocabulary.

Training employed bfloat16 precision with cosine scheduling on A40 GPUs, requiring approximately 72 hours total.

3.3. Supervised Fine-tuning

For instruction tuning, we curate 33,034 examples comprising Khasi translation pairs (10,097), English conversational data from Dolly-15k (15,000), and Khasi dialogue (7,937). Critical to success was removing 9,903 implicit translation examples lacking explicit instructions, which caused models to auto-translate all English inputs inappropriately.

We implement three essential technical fixes: (1) response-only loss masking to prevent instruction echoing, (2) mandatory EOS token verification to avoid infinite generation, and (3) consistent instruction formatting. Training uses LoRA rank 16 with learning rate $1e-4$ for 6 epochs, achieving final training loss of 0.85 in approximately 4 hours.

4. Results

Kren-M demonstrates strong functional performance on core tasks. For translation, the model correctly handles bidirectional Khasi-English translation with proper instruction following (e.g., "Translate to Khasi: How are you?" → "Kumno phi long?"). Critically, the model exhibits proper task-awareness, responding to English conversational inputs in English without unwanted auto-translation, while naturally conversing in Khasi when prompted in that language.

The model achieves clean generation with proper EOS token handling and no repetition loops. However, limitations include: (1) Garo vocabulary remains inactive pending dedicated training, (2) minor fluency gaps in complex English conversations typical of 2B-scale models, and (3) standard hallucination patterns on niche factual queries.

5. Discussion

Our work highlights several key insights for low-resource language modeling. Data quality proved paramount—the 45% loss reduction from Track B baseline (5.495) to our cleaned approach (2.974) demonstrates that meticulous cleaning outweighs raw corpus size. Multiple training failures due to contamination underscore the need for systematic quality control in low-resource settings.

Instruction tuning requires extreme precision. The removal of 9,903 implicit translation examples proved critical for maintaining task-awareness, while response-only loss masking and EOS token enforcement were essential for production-ready generation. These technical details, often glossed over in papers, have outsized impact on practical usability.

The two-stage CPT strategy (aggressive initial learning followed by conservative refinement) provided better control than single-pass training. However, Stage 1 alone proved sufficient for downstream tasks, suggesting the refinement stage may be optional for resource-constrained scenarios.

6. Conclusion

Kren-M is Meghalaya's first foundational AI model and the first production-ready bilingual LLM for the Khasi language.

Through custom tokenization, systematic data cleaning, and careful instruction tuning, we demonstrate that foundation models can be effectively adapted to truly low-resource languages. The model is publicly available on HuggingFace (MWirelabs/Kren-M), along with our cleaned corpus and training configurations. Future work includes activating Garo language capabilities, extending context length, and applying alignment techniques like DPO for improved response quality. This work provides a reproducible blueprint for developing language technologies for underserved linguistic communities.

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